CMEB Project 3

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11 June 2026

Contents

1 Point (1): Equations and boundary conditions

1.1 Device description

The device under study is a one-dimensional unipolar n^+-n-n^+ silicon structure of total length $L = 3\,\mu$m. The doping profile is smoothed as:

$$N_D(x) = N_D^{\text{low}} + \left(N_D^+ - N_D^{\text{low}}\right)\left(S_L(x) + S_R(x)\right), \tag{1}$$

where

$$S_L(x) = \frac{1}{2}\left(1 - \tanh\frac{x - L_c}{\sigma}\right), \qquad S_R(x) = \frac{1}{2}\left(1 + \tanh\frac{x - (L - L_c)}{\sigma}\right), \tag{2}$$

with $L_c = 0.45\,\mu$m and $\sigma = 30\,$nm. The doping values are $N_D^{\text{low}} = 10^{22}\,\text{m}^{-3}$ and $N_D^+ = 10^{24}\,\text{m}^{-3}$, giving a built-in potential scale $V_{bi} = V_T \log(N_D^+/N_D^{\text{low}}) \approx 0.119\,$V. The applied voltages range from $V_a = 0$ to $V_a = 0.5\,$V. The physical constants and material parameters used are summarized in Table 1.

Quantity	Symbol	Value
Elementary charge	q	1.602×10^{-19} C
Boltzmann constant	k_B	1.381×10^{-23} J/K
Lattice temperature	T_L	300 K
Thermal voltage	V_T	0.02585 V
Silicon permittivity	ε_{Si}	$11.7\,\varepsilon_0$
Low-field mobility	μ_n	0.1 $\text{m}^2/(\text{Vs})$
Energy relaxation time	τ_E	0.2 ps

Table 1: Physical constants and material parameters.

1.2 Equations solved

The non-isothermal unipolar model is formulated in the domain $\Omega = (0, L)$ and it is described as:

$$\begin{cases} -\dfrac{d}{dx}\left(\varepsilon\dfrac{d\varphi}{dx}\right) = q(N_D - n) & \text{(3a)} \\[2ex] \dfrac{dJ_n}{dx} = 0 & \text{(3b)} \\[2ex] J_n = q\mu_n\left(-n\dfrac{d\varphi}{dx} + \dfrac{d(n\vartheta)}{dx}\right) & \text{(3c)} \\[2ex] \dfrac{dS_n}{dx} = J_n E - \dfrac{3}{2}qn\dfrac{\vartheta - V_T}{\tau_E} & \text{(3d)} \\[2ex] S_n = -\lambda_n\dfrac{d\vartheta}{dx} - \dfrac{5}{2}\vartheta J_n & \text{(3e)} \end{cases}$$

where $E = -\frac{d\varphi}{dx}$ is the electric field, $V_T = \frac{k_B T_L}{q}$ is the lattice thermal voltage, ϑ is the electron thermal voltage and τ_E is the energy relaxation time.

The model seeks three unknowns: the electrostatic potential $\varphi(x)$, the electron density $n(x)$, and the electron thermal voltage $\vartheta(x) = k_B T_n(x)/q$, where T_n is the local electron temperature.

S_n is the electron energy flux, which is composed of two terms:

- thermal conductivity term

  $$\lambda_n \frac{d\vartheta}{dx}$$

  whose coefficient is

  $$\lambda_n = c_\lambda q\mu_n n\vartheta, \text{ with } c_\lambda = \frac{5}{2}. \tag{4}$$

- Convection term

  $$\frac{5}{2}\vartheta J_n$$

1.3 Boundary conditions

As boundary conditions, ohmic and thermally equilibrated contacts are imposed at $x = 0$ and $x = L$:

$$\begin{cases} \varphi(0) = 0, & \varphi(L) = V_a, \\ n(0) = N_D^+, & n(L) = N_D^+, \\ \vartheta(0) = V_T, & \vartheta(L) = V_T. \end{cases} \tag{5}$$

The last condition enforces that electrons thermalize completely at the contacts, i.e. $T_n = T_L$ at $x = 0$ and $x = L$.

2 Point (2): Discrete SG fluxes

2.1 Mesh and edge notation

The domain $\Omega = (0, L)$ is discretized on a uniform mesh of N nodes:

$$x_i = (i-1)h, \quad i = 1, \ldots, N, \qquad h = \frac{L}{N-1}. \tag{6}$$

2.2 Non-isothermal Scharfetter–Gummel current flux

For fixed φ and ϑ, the electron current on edge $[x_i, x_{i+1}]$ is approximated by the non-isothermal Scharfetter–Gummel flux:

$$J_{n,i+\frac{1}{2}} = \frac{q\mu_n \vartheta^{lm}_{i+\frac{1}{2}}}{h} \left[n_{i+1} B(\eta_{i+\frac{1}{2}}) - n_i B(-\eta_{i+\frac{1}{2}}) \right], \tag{7}$$

where the driving force is

$$\eta_{i+\frac{1}{2}} = \frac{(\varphi_{i+1} - \varphi_i) - (\vartheta_{i+1} - \vartheta_i)}{\vartheta^{lm}_{i+\frac{1}{2}}}, \tag{8}$$

and $\vartheta^{lm}_{i+\frac{1}{2}} = \mathcal{L}(\vartheta_i, \vartheta_{i+1})$ is the logarithmic mean:

$$\mathcal{L}(a, b) = \begin{cases} \dfrac{b-a}{\log b - \log a} & a \neq b, \\ a & a = b. \end{cases} \tag{9}$$

The Bernoulli function is defined as:

$$B(z) = \frac{z}{e^z - 1}, \qquad B(0) = 1, \tag{10}$$

and it is evaluated via Taylor expansion for $|z| \ll 1$.

In the isothermal limit $\vartheta_i = \vartheta_{i+1} = V_T$, equation (7) reduces to the standard SG flux. The continuity equation is enforced as:

$$J_{n,i+\frac{1}{2}} - J_{n,i-\frac{1}{2}} = 0, \qquad i = 2, \ldots, N-1. \tag{11}$$

The mobility $\mu_n = 0.1\ \mathrm{m}^2/(\mathrm{Vs})$ is kept constant throughout the domain. For fixed φ and ϑ, this is a linear system for the interior values of n.

Substituting the non-isothermal SG flux (7) into the discrete continuity equation $J_{n,i+\frac{1}{2}} - J_{n,i-\frac{1}{2}} = 0$ for the k-th iteration of the Gummel scheme, we get:

$$\frac{q\mu_n \vartheta^{lm}_{i+\frac{1}{2}}}{h} \left[n_{i+1} B\left(\eta_{i+\frac{1}{2}}\right) - n_i B\left(-\eta_{i+\frac{1}{2}}\right) \right] - \frac{q\mu \vartheta^{lm}_{i-\frac{1}{2}}}{h} \left[n_i B\left(\eta_{i-\frac{1}{2}}\right) - n_{i-1} B\left(-\eta_{i-\frac{1}{2}}\right) \right] = 0 \tag{12}$$

Collecting terms in n_{i-1}, n_i, n_{i+1} yields to:

$$n_{i-1} \left[\vartheta^{lm}_{i-\frac{1}{2}} B\left(-\eta_{i-\frac{1}{2}}\right) \right] + n_i \left[-\vartheta^{lm}_{i+\frac{1}{2}} B\left(-\eta_{i+\frac{1}{2}}\right) - \vartheta^{lm}_{i-\frac{1}{2}} B\left(\eta_{i-\frac{1}{2}}\right) \right] + n_{i+1} \left[\vartheta^{lm}_{i+\frac{1}{2}} B\left(\eta_{i+\frac{1}{2}}\right) \right] = 0 \tag{13}$$

This equation can be solved as a linear system $An = b$ using a tridiagonal matrix with the following structure:

$$\text{subdiagonal:}\quad a_{sub} = -\frac{q\mu_n\vartheta^{lm}_{i+\frac{1}{2}}}{h}B\Big(-\eta_{i+\frac{1}{2}}\Big)\,, \tag{14}$$

$$\text{diagonal:}\quad a_{main} = \frac{q\mu_n}{h}\left[\vartheta^{lm}_{i+\frac{1}{2}}B\Big(-\eta_{i+\frac{1}{2}}\Big) + \vartheta^{lm}_{i-\frac{1}{2}}B\Big(\eta_{i-\frac{1}{2}}\Big)\right], \tag{15}$$

$$\text{superdiagonal:}\quad a_{top} = -\frac{q\mu_n\vartheta^{lm}_{i+\frac{1}{2}}}{h}B\Big(\eta_{i+\frac{1}{2}}\Big)\,. \tag{16}$$

The RHS of the system

$$b = \begin{bmatrix} \frac{q\mu_n\vartheta^{lm}_{i+\frac{1}{2}}}{h}B\Big(-\eta_{i+\frac{1}{2}}\Big) \\ 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \\ \frac{q\mu_n\vartheta^{lm}_{i+\frac{1}{2}}}{h}B\Big(\eta_{i+\frac{1}{2}}\Big) \end{bmatrix} \tag{17}$$

2.3 Scharfetter-Gummel energy flux

The edge thermal conductivity is computed from arithmetic averages:

$$\lambda_{i+\frac{1}{2}} = c_\lambda q\mu_n n^{\text{avg}}_{i+\frac{1}{2}}\vartheta^{\text{avg}}_{i+\frac{1}{2}}, \qquad n^{\text{avg}}_{i+\frac{1}{2}} = \frac{n_i + n_{i+1}}{2}, \qquad \vartheta^{\text{avg}}_{i+\frac{1}{2}} = \frac{\vartheta_i + \vartheta_{i+1}}{2}. \tag{18}$$

The energy Péclet number on each edge is:

$$P^E_{i+\frac{1}{2}} = \frac{5J_{n,i+\frac{1}{2}}h}{2\lambda_{i+\frac{1}{2}}}. \tag{19}$$

The exponentially fitted energy flux is:

$$S_{n,i+\frac{1}{2}} = \frac{\lambda_{i+\frac{1}{2}}}{h}\left[\vartheta_i B\Big(-P^E_{i+\frac{1}{2}}\Big) - \vartheta_{i+1}B\Big(P^E_{i+\frac{1}{2}}\Big)\right]. \tag{20}$$

For $J_n = 0$, $P^E = 0$ and (20) reduces to the centered Fourier law:

$$S_{n,i+\frac{1}{2}} = -\lambda_{i+\frac{1}{2}}\frac{\vartheta_{i+1} - \vartheta_i}{h}. \tag{21}$$

The discrete energy balance at each interior node reads:

$$S_{n,i+\frac{1}{2}} - S_{n,i-\frac{1}{2}} = h\left[H_i - \alpha_i(\vartheta_i - V_T)\right], \qquad i = 2, \ldots, N-1, \tag{22}$$

where the nodal Joule heating and relaxation coefficients are:

$$H_i = \frac{1}{2}\left(J_{n,i+\frac{1}{2}}E_{i+\frac{1}{2}} + J_{n,i-\frac{1}{2}}E_{i-\frac{1}{2}}\right), \qquad \alpha_i = \frac{3}{2}\frac{qn_i}{\tau_E}, \qquad E_{i+\frac{1}{2}} = -\frac{\varphi_{i+1} - \varphi_i}{h}. \tag{23}$$

For fixed n, φ, J_n, and frozen $\lambda_{i+\frac{1}{2}}$, this is a linear system ($A\vartheta = b$) for the interior values of ϑ. Solving (22) we get:

$$\frac{\lambda_{i+\frac{1}{2}}}{h}\left[\vartheta_i B\Big(-P^E_{i+\frac{1}{2}}\Big) - \vartheta_{i+1}B\Big(P^E_{i+\frac{1}{2}}\Big)\right] - \frac{\lambda_{i-\frac{1}{2}}}{h}\left[\vartheta_{i-1}B\Big(-P^E_{i-\frac{1}{2}}\Big) - \vartheta_i B\Big(P^E_{i-\frac{1}{2}}\Big)\right] = h\left[H_i - \alpha_i(\vartheta_i - V_T)\right] \tag{24}$$

Collecting the terms ϑ_{i-1}, ϑ_i and ϑ_{i+1}:

$$
\begin{aligned}
&-\vartheta_{i-1}\left[\frac{\lambda_{i-\frac{1}{2}}}{h}B\Big(-P^E_{i-\frac{1}{2}}\Big)\right]+\vartheta_i\left[\frac{\lambda_{i+\frac{1}{2}}}{h}B\Big(-P^E_{i+\frac{1}{2}}\Big)+\frac{\lambda_{i-\frac{1}{2}}}{h}B\Big(P^E_{i-\frac{1}{2}}\Big)+h\cdot\alpha_i\right]\\
&-\vartheta_{i+1}\left[\frac{\lambda_{i+\frac{1}{2}}}{h}B\Big(P^E_{i+\frac{1}{2}}\Big)\right]=h\left(H_i+\alpha_i\cdot V_T\right)
\end{aligned}
\tag{25}
$$

This equation can be solved using a tridiagonal matrix composed as:

$$
\text{subdiagonal:}\quad a_{sub}=-\frac{\lambda_{i+\frac{1}{2}}}{h}B\Big(-P^E_{i+\frac{1}{2}}\Big),
\tag{26}
$$

$$
\text{diagonal:}\quad a_{main}=\frac{\lambda_{i+\frac{1}{2}}}{h}B\Big(-P^E_{i+\frac{1}{2}}\Big)+\frac{\lambda_{i-\frac{1}{2}}}{h}B\Big(P^E_{i-\frac{1}{2}}\Big)+h\cdot\alpha_i,
\tag{27}
$$

$$
\text{superdiagonal:}\quad a_{top}=-\frac{\lambda_{i+\frac{1}{2}}}{h}B\Big(P^E_{i+\frac{1}{2}}\Big),
\tag{28}
$$

and the right-hand side vector:

$$
b_i=h\left(H_i+\alpha_i\cdot V_T\right).
\tag{29}
$$

The edges of the RHS vector have to include the boundary conditions:

$$
b(1)=h\left(H_1+\alpha_1\cdot V_T\right)+\frac{\lambda_{\frac{3}{2}}}{h}B\Big(-P^E_{\frac{3}{2}}\Big)\qquad b(N)=h\left(H_N+\alpha_N\cdot V_T\right)+\frac{\lambda_{N+\frac{1}{2}}}{h}B\Big(P^E_{N+\frac{1}{2}}\Big),
\tag{30}
$$

3 Point (3): Gummel iteration

The nonlinear coupled system (3) is solved by a Gummel iteration, which decouples the equations and solves them sequentially at each step. For each applied voltage V_a, the iteration proceeds as follows.

1. **Initialization.** Two initialization strategies were considered:

   a. **Isothermal initialization (I).** For every bias V_a, the converged isothermal solution at the same voltage is used as initial guess for φ and n, while ϑ is set to V_T uniformly for $V_a = 0$ and taken from the non-isothermal solution at the previous bias for $V_a > 0$.

   b. **Non-isothermal continuation (C).** For $V_a = 0$, the quasi-neutral equilibrium is used as an initial guess: $n = N_D(x)$, $\varphi = V_T \log(N_D/N_D^+) + V_a x/L$, and $\vartheta = V_T$ uniformly. For $V_a > 0$, all three fields are initialized from the converged non-isothermal solution at the previous bias, with an affine correction $\Delta V_a \cdot x/L$ applied to φ.

   Both strategies were tested and found to converge to identical solutions within the prescribed tolerance, confirming the uniqueness and robustness of the non-isothermal solution. Strategy (C) is adopted throughout this work.

2. **Continuity equation.** Given φ^k and ϑ^k, solve the non-isothermal SG continuity equation (11) for n^{k+1}, which is found by computing (13) as:

$$
n^{k+1}_{i-1}\left[\vartheta^{lm,k}_{i-\frac{1}{2}}B\left(-\eta^k_{i-\frac{1}{2}}\right)\right]+n^{k+1}_i\left[-\vartheta^{lm,k}_{i+\frac{1}{2}}B\left(-\eta^k_{i+\frac{1}{2}}\right)-\vartheta^{lm,k}_{i-\frac{1}{2}}B\left(\eta^k_{i-\frac{1}{2}}\right)\right]+n^{k+1}_{i+1}\left[\vartheta^{lm,k}_{i+\frac{1}{2}}B\left(\eta^k_{i+\frac{1}{2}}\right)\right]=0
\tag{31}
$$

3. **Current update.** Compute the edge currents J_n^{k+1} using (7) with the updated n^{k+1}.

$$
J^{k+1}_{n,i+\frac{1}{2}}=\frac{q\mu_n\vartheta^{lm,k}_{i+\frac{1}{2}}}{h}\left[n^{k+1}_{i+1}B(\eta^k_{i+\frac{1}{2}})-n^{k+1}_iB(-\eta^k_{i+\frac{1}{2}})\right],
\tag{32}
$$

4. **Energy equation.** Given n^{k+1}, φ^k, and J_n^{k+1}, solve the discrete energy balance (22) for ϑ, with boundary conditions $\vartheta_1 = \vartheta_N = V_T$. The terms needed to solve the equation are computed as follows:

   - The edge conductivity λ^k is computed from n^{k+1} and the previous ϑ^k according to (18)

- $P^{E,k}$ is computed from λ^k and J_n^{k+1} according to (19)
- α^k is computed from n^{k+1} according to (23)
- H^k is computed from J^{k+1} and E^k according to (23)

Evaluating (25) for the current Gummel iteration:

$$
\begin{aligned}
&-\vartheta_{i-1}^{k+1}\left[\frac{\lambda_{i-\frac{1}{2}}^k}{h}B\Big(-P_{i-\frac{1}{2}}^{E,k}\Big)\right]+\vartheta_i^{k+1}\left[\frac{\lambda_{i+\frac{1}{2}}^k}{h}B\Big(-P_{i+\frac{1}{2}}^{E,k}\Big)+\frac{\lambda_{i-\frac{1}{2}}^k}{h}B\Big(P_{i-\frac{1}{2}}^{E,k}\Big)+h\cdot\alpha_i^k\right]\\
&-\vartheta_{i+1}^{k+1}\left[\frac{\lambda_{i+\frac{1}{2}}^k}{h}B\Big(P_{i+\frac{1}{2}}^{E,k}\Big)\right]=h\left(H_i^k+\alpha_i^k\cdot V_T\right)
\end{aligned}
\tag{33}
$$

so that the energy subproblem reduces to a *linear* tridiagonal system for the interior values of ϑ. In this project, the term $h\cdot\alpha_i^k$ has been used with ϑ^{k+1} and included as a mass matrix term to make the matrix associated to the linear system more stable.

5. **Poisson equation.** Solve the Gummel-linearized Poisson equation for φ^{k+1}. Using the local linearization $\partial n/\partial\varphi \simeq n/\vartheta$, the discrete system reads:

$$
-\varepsilon\frac{\varphi_{i+1}^{k+1}-2\varphi_i^{k+1}+\varphi_{i-1}^{k+1}}{h^2}+\frac{qn_i^k}{\vartheta_i^k}\varphi_i^{k+1}=q(N_{D,i}-n_i^k)+\frac{qn_i^k}{\vartheta_i^k}\varphi_i^k \tag{34}
$$

for $i=2,\ldots,N-1$, with Dirichlet conditions $\varphi_1=0$ and $\varphi_N=V_a$. This leads to an associated linear system.

6. **Damping.** After each Gummel step, the updated fields are blended with the previous iterate:

$$
n\leftarrow(1-\omega_n)n^k+\omega_n n,\qquad \vartheta\leftarrow(1-\omega_\vartheta)\vartheta^k+\omega_\vartheta\vartheta,\qquad \varphi\leftarrow(1-\omega_\varphi)\varphi^k+\omega_\varphi\varphi. \tag{35}
$$

No damping was required for the voltage range considered, so $\omega_n=\omega_\varphi=\omega_\vartheta=1$ throughout. Convergence with reduced damping ($\omega=0.5$) was verified to yield identical results with a larger number of iterations.

7. **Convergence check.** The iteration is stopped when:

$$
\max\left\{\frac{\|\varphi-\varphi^k\|_\infty}{\max(V_T,\,\|\varphi\|_\infty)},\ \frac{\|n-n^k\|_\infty}{\|n\|_\infty},\ \frac{\|\vartheta-\vartheta^k\|_\infty}{\|\vartheta\|_\infty}\right\}<10^{-10}. \tag{36}
$$

The maximum number of iterations is set to 500.

4 Point (4): Validation at $Va=0$ + baseline comparison

4.1 Validation at $Va=0$

At zero applied bias ($V_a=0$), the device must recover the thermal equilibrium solution. This represents a fundamental consistency check for the non-isothermal model. Since no external driving force is applied, the electron temperature must coincide with the lattice temperature T_L, and the system must reduce to the isothermal equilibrium.

4.1.1 Expected equilibrium conditions

At thermal equilibrium, the following conditions must hold simultaneously:

$$
J_n\simeq 0, \tag{37}
$$

$$
T_n(x)\simeq T_L,\quad \text{i.e.,}\quad \vartheta(x)\simeq V_T, \tag{38}
$$

$$
n(x)\simeq N_D(x), \tag{39}
$$

$$
\varphi(x)\simeq V_T\log\left(\frac{N_D(x)}{N_D^+}\right). \tag{40}
$$

The first condition expresses current conservation with no net carrier flow. The second states that, without Joule heating, the electron gas thermalizes with the lattice. The third and fourth are the classical equilibrium relations for the carrier density and the built-in potential.

4.1.2 Numerical results

Figure 1 shows the computed profiles of $n(x)$, $T_n(x)$, and J_n at $V_a = 0$ for both the isothermal and non-isothermal models.

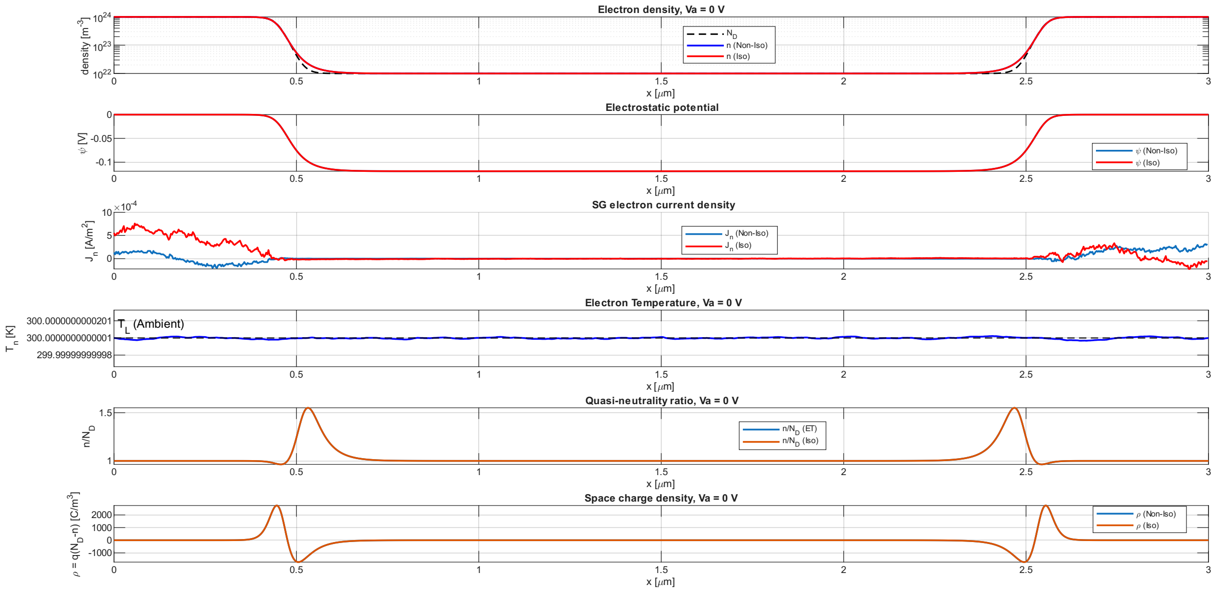

Figure 1: Zero-bias comparison

It can be noticed that the electron temperature is very close to the lattice temperature (300K), as expected. The non-isothermal schemes have been executed with a tolerance of 10^{-10} and without any damping. Furthermore, to make a fair comparison, the non-isothermal scheme has been executed with the "non-isothermal continuation" boundary condition shown in Section 3. The scheme converges in 6 iterations.

In Table 2, some key results are reported:

Quantity	Isothermal	Non-isothermal
$\overline{J}_n$ [A/m^2]	7.15×10^{-5}	4.57×10^{-5}
$\bar{\vartheta}$ [V]	$V_T = 0.02585$	0.02590
$\varphi(L/2)$ [V]	-0.1191	-0.1191
$\varphi_{\text{built-in}}$ [V]	-0.1191	-0.1191

Table 2: Zero-bias validation ($V_a = 0$): comparison between isothermal and non-isothermal models.

Where:

- $\overline{J}_n$ is the average current flux
- $\bar{\vartheta}$ is the average electron voltage temperature
- $\varphi(L/2)$ is the potential at the center of the device

4.1.3 Interpretation

The agreement between the non-isothermal model and the known equilibrium solution confirms that:

1. the current flux correctly reduces to zero net current at equilibrium;
2. the inclusion of the energy model leads to the trivial result of $\vartheta \approx V_T$ as expected
3. the Gummel-linearized Poisson equation (34) leads to φ equal to the built-in potential correctly.

This zero-bias check must be passed before any biased simulation is considered reliable.

4.2 Baseline comparison + Point (6): T_n plots

In the following figures, from 2 to 7, the comparative plots for the V_a required are presented:

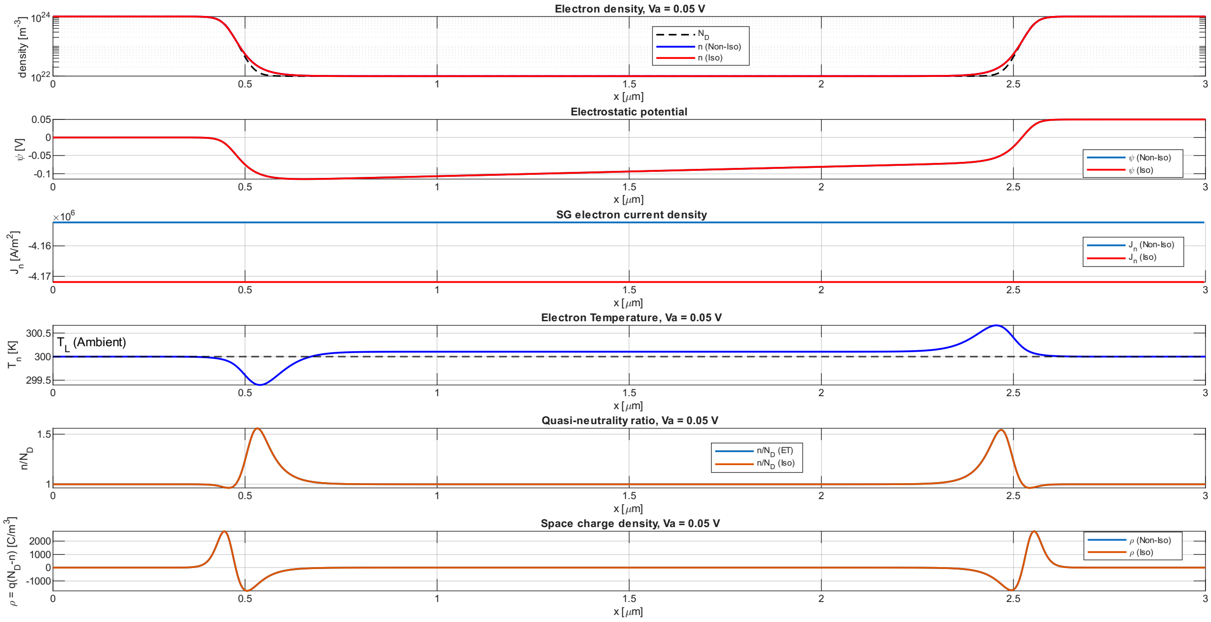

Figure 2: Comparison for $V_a = 0.05$ V.

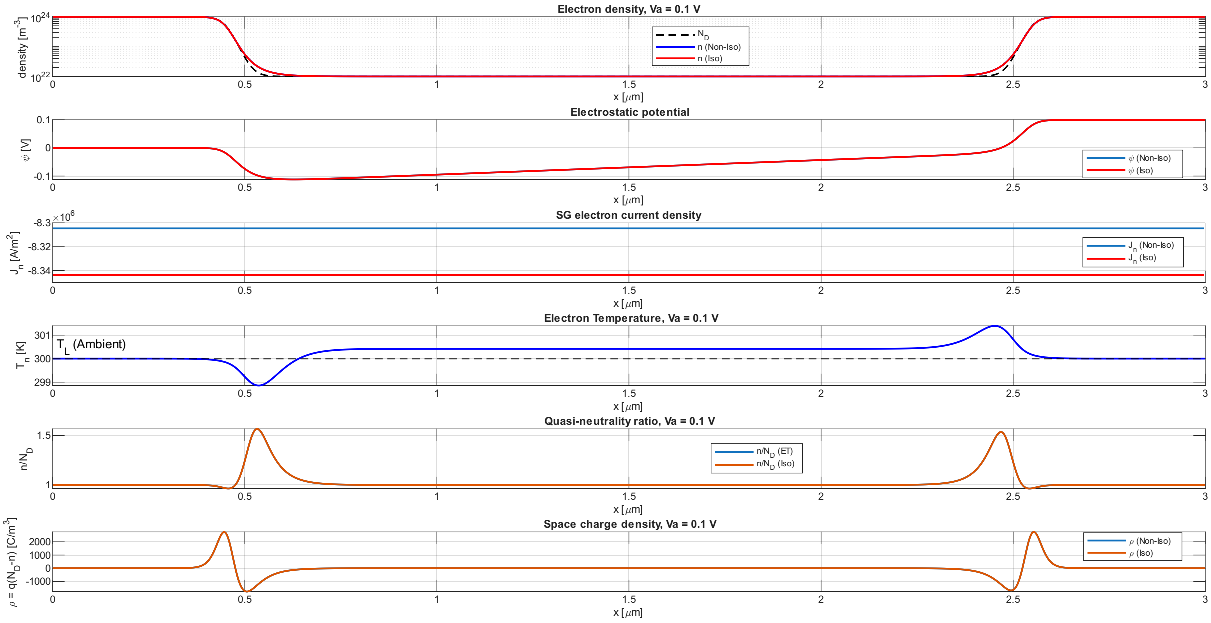

Figure 3: Comparison for $V_a = 0.1$ V.

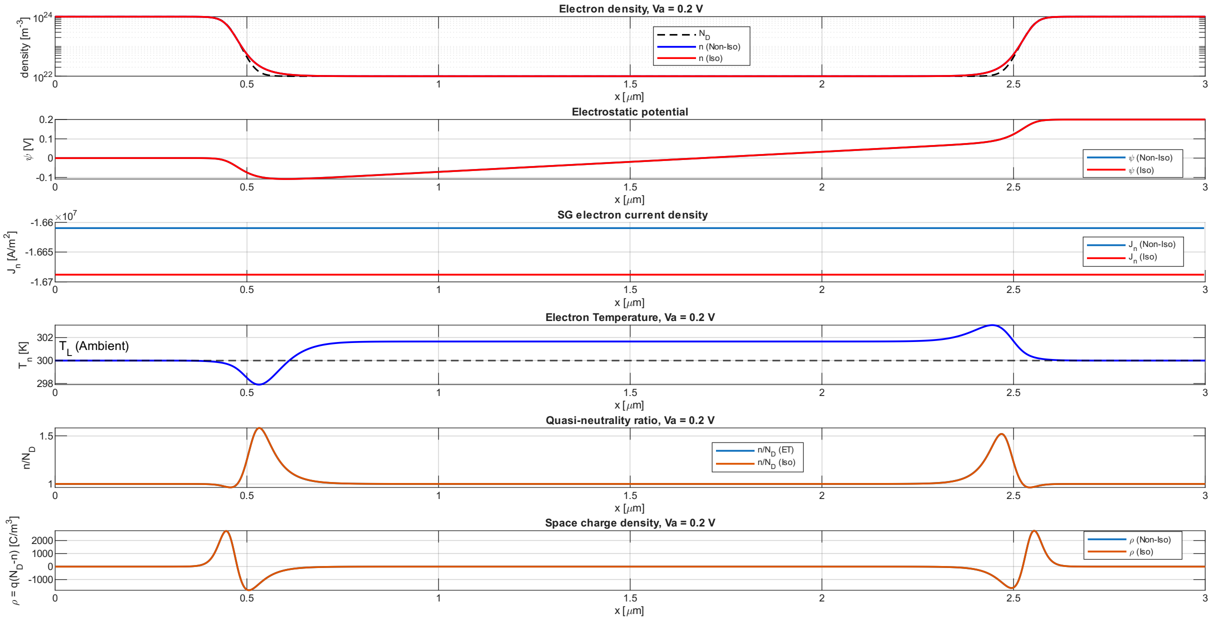

Figure 4: Comparison for $V_a = 0.2$ V.

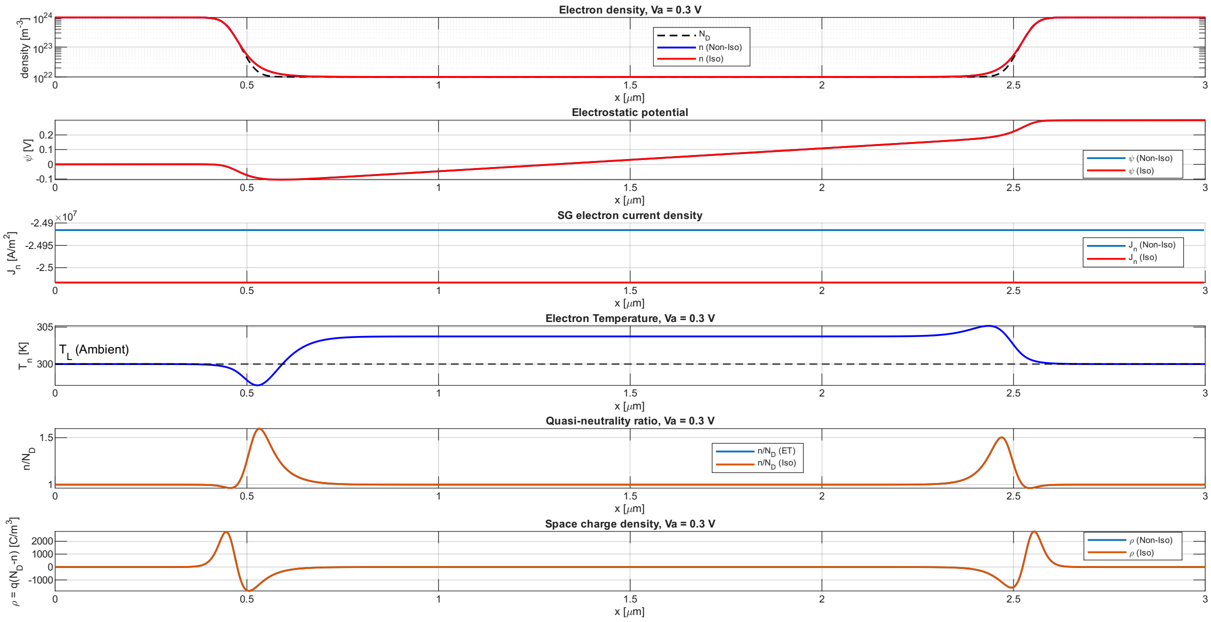

Figure 5: Comparison for $V_a = 0.3$ V.

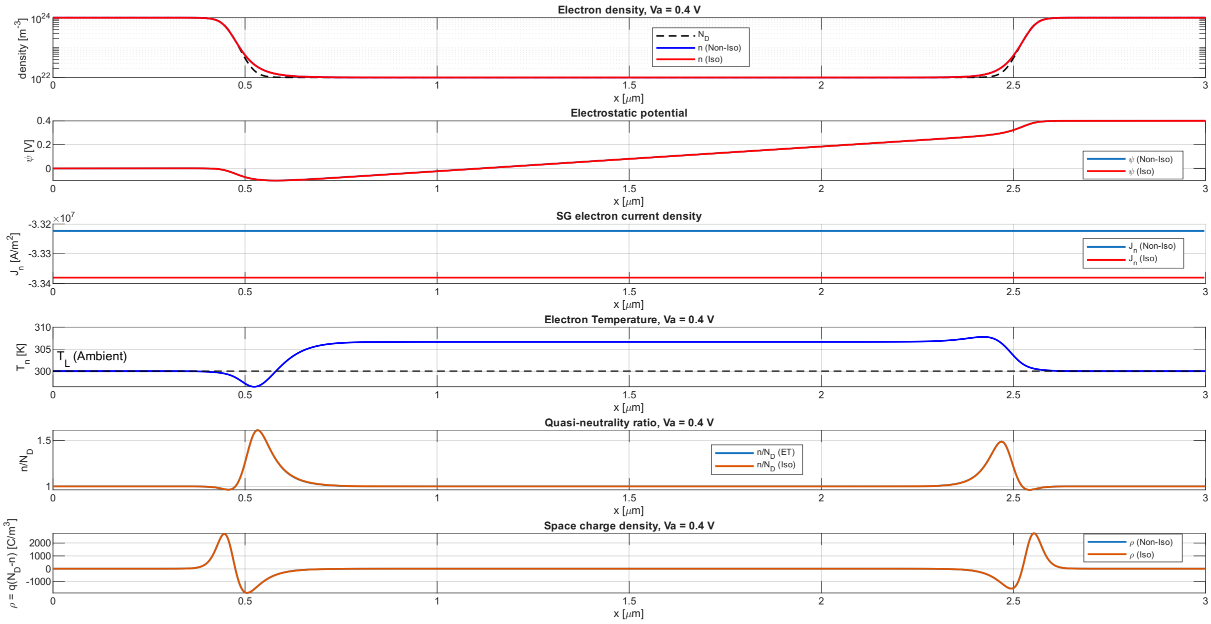

Figure 6: Comparison for $V_a = 0.4$ V.

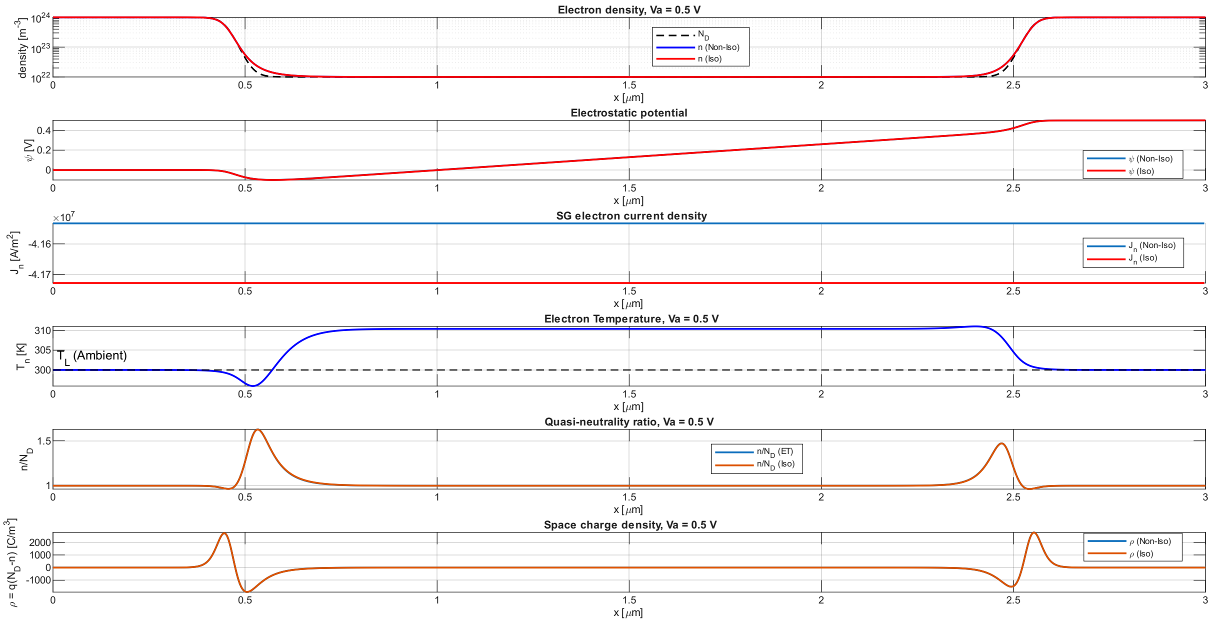

Figure 7: Comparison for $V_a = 0.5$ V.

The non-isothermal model is computed with $\tau_E = 0.2$ ps. There are no major differences between the isothermal and non-isothermal models.

The spatial profiles of the electrostatic potential ψ, electron concentration n, and space charge density ρ exhibit negligible discrepancies when switching from the isothermal Drift-Diffusion model to the non-isothermal Energy Transport framework. This behavior can be easily explained by looking at the variation in the temperature of the electrons, which is not high enough to cause large discrepancies.

In contrast to the electrostatic quantities, the electron current density J_n shows a measurable, albeit small, variation between the two models.
For instance, at $V_a = 0.5$ V, the electron temperature $T_n(x)$ reaches a peak of approximately 310 K in the low-doped n^- region, where the electric field is strongest. This local heating introduces an additional diffusion-like contribution to the current through the gradient of the electron thermal voltage ϑ:

$$J_n = q\mu_n \left(-n\frac{d\varphi}{dx} + \frac{d(n\vartheta)}{dx} \right) = J_n^{\text{iso}} + q\mu_n \frac{d(n\vartheta)}{dx} \tag{41}$$

Since $\vartheta > V_T$ in the n^- region, the diffusion term $q\mu_n \frac{d(n\vartheta)}{dx}$ is enhanced with respect to the isothermal case. However, the diffusion contribution opposes the drift term $-q\mu_n n \frac{d\varphi}{dx}$: in the n^- region, where ϑ is approximately uniform and elevated, the net effect is a partial cancellation of the drift current, yielding $|J_{n,\text{non-iso}}| < |J_{n,\text{iso}}|$. This is consistent with the data in Table 3, where a relative reduction of approximately 0.47% is observed across the full bias range.

The electron temperature profile $T_n(x)$ exhibits a downward peak at the first junction: it is a consequence of energy conservation. At the first junction ($x = 0.5\mu m$), the electrons gain potential energy due to the built-in potential, which slows them down, reducing their kinetic energy and temperature.
Same reasoning can be applied analogously at the second junction ($x = 2.5\mu m$), where the built-in potential accelerates the electrons and slightly increases their temperature.

5 Point (5): current-voltage plots for the isothermal and non-isothermal models

Figure 8 shows the current-voltage characteristic for both the isothermal drift-diffusion model and the non-isothermal energy-balance model over the bias range $V_a \in [0, 0.50]$ V with $\tau_E = 0.2$ps.

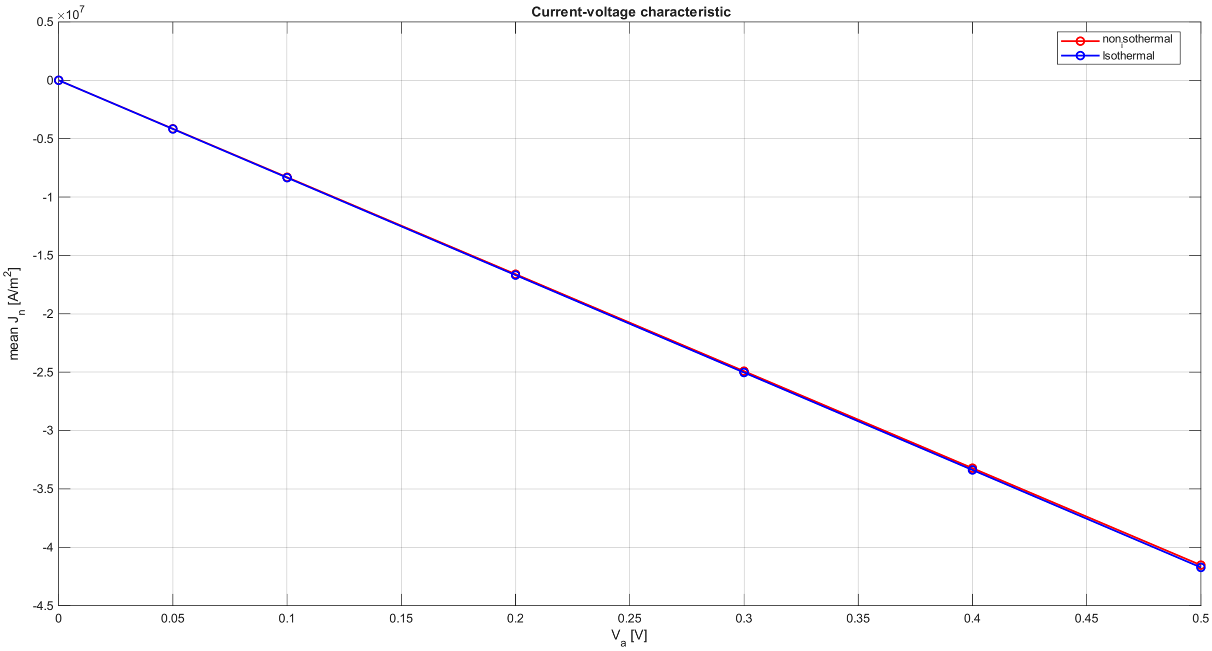

Figure 8: Current-voltage characteristic: comparison between the isothermal and non-isothermal models for $V_a \in [0, 0.50]$ V.

V_a [V]	$\lvert\Delta J_n\rvert$ [A/m^2]	$\lvert\Delta J_n\rvert/\lvert J_{n,\text{ISO}}\rvert$ [%]
0.05	1.958×10^4	0.4693
0.10	3.915×10^4	0.4692
0.20	7.826×10^4	0.4689
0.30	1.1726×10^5	0.4684
0.40	1.5612×10^5	0.4677
0.50	1.9480×10^5	0.4668

Table 3: Absolute and relative deviation of the non-isothermal current from the isothermal baseline.

The two curves are nearly indistinguishable on the scale of the plot, with a relative deviation of approximately 0.47% across the entire bias range (see Table 3, $|\Delta J_n| = J_{n,iso} - J_{n,non-iso}$). This result is physically expected: at the applied voltages considered here, the electron temperature rises only moderately above the lattice temperature — reaching a maximum of approximately 310 K at $V_a = 0.50$ V

— so that the additional diffusion-like contribution $q\mu_n n \, \mathrm{d}\vartheta/\mathrm{d}x$ to the current remains a small correction to the dominant drift term.

Furthermore, in this low-field regime, the velocity ratio $\max_x |u_n|/v_{\mathrm{sat}} \ll 1$ (see Table 4), confirming that the electron gas is far from saturation and that the isothermal model remains a good approximation for the current. The velocity ratio $\max_x |u_n|/v_{\mathrm{sat}}$ grows monotonically with the applied bias, as expected from the increasing electric field in the n^- region.
For the largest voltage $V_a = 0.50$ V, the ratio approaches the threshold of 0.3, above which the constant-mobility approximation becomes questionable. The constant-mobility model assumes a linear drift velocity $u_n = \mu_n E$, whereas the physical electron velocity in silicon saturates as:

$$u_n = \frac{\mu_n E}{1 + \mu_n |E|/v_{\mathrm{sat}}}. \tag{42}$$

Expanding to first order in $\mu_n|E|/v_{\mathrm{sat}} = |u_n|/v_{\mathrm{sat}}$:

$$u_n \approx \mu_n E \left(1 - \frac{|u_n|}{v_{\mathrm{sat}}} + \mathcal{O}\left(\frac{|u_n|^2}{v_{\mathrm{sat}}^2}\right)\right), \tag{43}$$

Therefore, a velocity ratio of $|u_n|/v_{\mathrm{sat}} = 0.3$ corresponds directly to a 30% error on the carrier velocity, and consequently on the current density $J_n = -qnu_n$.

V_a [V]	$\max_x T_n$ [K]	$\max_x (T_n - T_L)/T_L$	$\max_x |u_n|/v_{\mathrm{sat}}$
0.05	300.66	2.21×10^{-3}	2.59×10^{-2}
0.10	301.39	4.64×10^{-3}	5.18×10^{-2}
0.20	303.09	1.03×10^{-2}	1.04×10^{-1}
0.30	305.18	1.73×10^{-2}	1.56×10^{-1}
0.40	307.77	2.59×10^{-2}	2.07×10^{-1}
0.50	311.02	3.67×10^{-2}	2.59×10^{-1}

Table 4: Maximum electron temperature and velocity ratio across the bias range. The low-field threshold is $|u_n|/v_{\mathrm{sat}} < 0.2$.

6 Point (7): Energy residual and conservation

In Table 5, the energy diagnostic have been reported:

V_a **[V]**	**Energy Normalized Residual**	**Relative Imbalance** (Global Balance)
0.00	6.2296×10^{0}	5.2570×10^{0}
0.05	2.7554×10^{-10}	1.5100×10^{-6}
0.10	1.4542×10^{-10}	1.5128×10^{-6}
0.20	6.2774×10^{-11}	1.5130×10^{-6}
0.30	7.0592×10^{-11}	1.5131×10^{-6}
0.40	3.6730×10^{-11}	1.5132×10^{-6}
0.50	2.9765×10^{-11}	1.5134×10^{-6}

Table 5: Energy residual and global energy balance diagnostics for different bias points V_a.

The energy residual computed is defined as:

$$R_i^E = \frac{S_{n,i+\frac{1}{2}} - S_{n,i-\frac{1}{2}}}{h} - [H_i - \alpha_i(\vartheta_i - V_T)], \qquad i = 2, \ldots, N-1, \tag{44}$$

and is reported normalized as:

$$\frac{\|R^E\|_\infty}{\max_i \left(|H_i| + |\alpha_i(\vartheta_i - V_T)|\right) + \varepsilon_{mach}}. \tag{45}$$

The global energy balance identity

$$\int_0^L J_n E\,dx = S_n(L) - S_n(0) + \int_0^L \frac{3}{2} qn \frac{\vartheta - V_T}{\tau_E}\,dx \tag{46}$$

is verified numerically by computing each term separately and reporting the relative imbalance:

$$\text{imbalance} = \frac{|\text{LHS} - \text{RHS}|}{|\text{LHS}| + \varepsilon_{mach}}. \tag{47}$$

It can be noticed that the diagnostics have been respected for every bias point but $V_a = 0$. At $V_a = 0$, the energy residual and global energy balance diagnostics show large values ($\|R^E\|/\text{denom} \approx 6.2$ and relative imbalance ≈ 5.3). This is expected: at equilibrium $J_n \approx 0$ and $\vartheta \approx V_T$ uniformly, so both the Joule heating H_i and the relaxation term $\alpha_i(\vartheta_i - V_T)$ are nearly zero. The denominator of the normalized residual approaches zero, making the ratio ill-conditioned and not meaningful as an indicator.

For all $V_a > 0$, both diagnostics recover values well below 10^{-9}, confirming the correctness of the discretization.

7 Point (8): relaxation time variation

In order to test the model for different τ_E, it has been tested with two different voltage biases: $V_a = 0.2$V and $V_a = 0.5$V. These two values offer a good insight into what happens at moderate bias and at higher bias. Table 6 shows the results obtained:

τ_E **[ps]**	V_a **[V]**	$\max(T_n)$ **[K]**	$\max((T_n - T_L)/T_L)$	**mean**(J_n) **[A/m^2]**
0.05	0.2	300.95	3.1629×10^{-3}	-1.6662×10^7
	0.5	303.05	1.0169×10^{-2}	-4.1663×10^7
0.10	0.2	301.75	5.8334×10^{-3}	-1.6642×10^7
	0.5	305.85	1.9513×10^{-2}	-4.1611×10^7
0.20	0.2	303.09	1.0301×10^{-2}	-1.6610×10^7
	0.5	311.02	3.6746×10^{-2}	-4.1533×10^7
0.50	0.2	306.10	2.0334×10^{-2}	-1.6548×10^7
	0.5	325.82	8.6064×10^{-2}	-4.1393×10^7
1.00	0.2	309.81	3.2716×10^{-2}	-1.6489×10^7
	0.5	351.38	1.7125×10^{-1}	-4.1287×10^7

Table 6: Simulation results for different energy relaxation times τ_E and bias points V_a, showing maximum electron temperature, maximum relative temperature variation, and average current density.

It can be noticed that the electron temperature increases for a larger relaxation time, as expected. A larger relaxation time means lower energy transfer from the electrons to the lattice via scattering.

Furthermore, the current reduces for larger τ_E, consistently with Section 5.

8 Point (9): mesh refinement

Different meshes ($N = 500$, $N = 1000$, $N = 1500$)have been tested on the model for two different biases ($V_a = 0.2$V and $V_a = 0.5$V) and with $\tau_E = 1$ps.

In Table 7 the residuals and the temperature are shown, while in Figure 9 and Figure 10, the plots of the currents along the x-axis are shown.

N	$\|R^P\|_\infty/(qN_D^+)$	Normalized Energy Residual	Rel. imbalance	$\max(T_n)$ [K]
		$V_a = 0.2\,\mathrm{V}$		
500	6.756×10^{-12}	2.464×10^{-11}	3.000×10^{-6}	309.812
1000	6.762×10^{-12}	9.529×10^{-11}	7.635×10^{-7}	309.815
1500	6.770×10^{-12}	1.856×10^{-10}	3.414×10^{-7}	309.816
		$V_a = 0.5\,\mathrm{V}$		
500	1.442×10^{-11}	8.707×10^{-12}	3.030×10^{-6}	351.383
1000	1.455×10^{-11}	3.866×10^{-11}	7.709×10^{-7}	351.376
1500	1.457×10^{-11}	7.188×10^{-11}	3.447×10^{-7}	351.375

Table 7: Mesh refinement study at $V_a = 0.2\,\mathrm{V}$ and $V_a = 0.5\,\mathrm{V}$: residuals and maximum electron temperature for three mesh sizes.

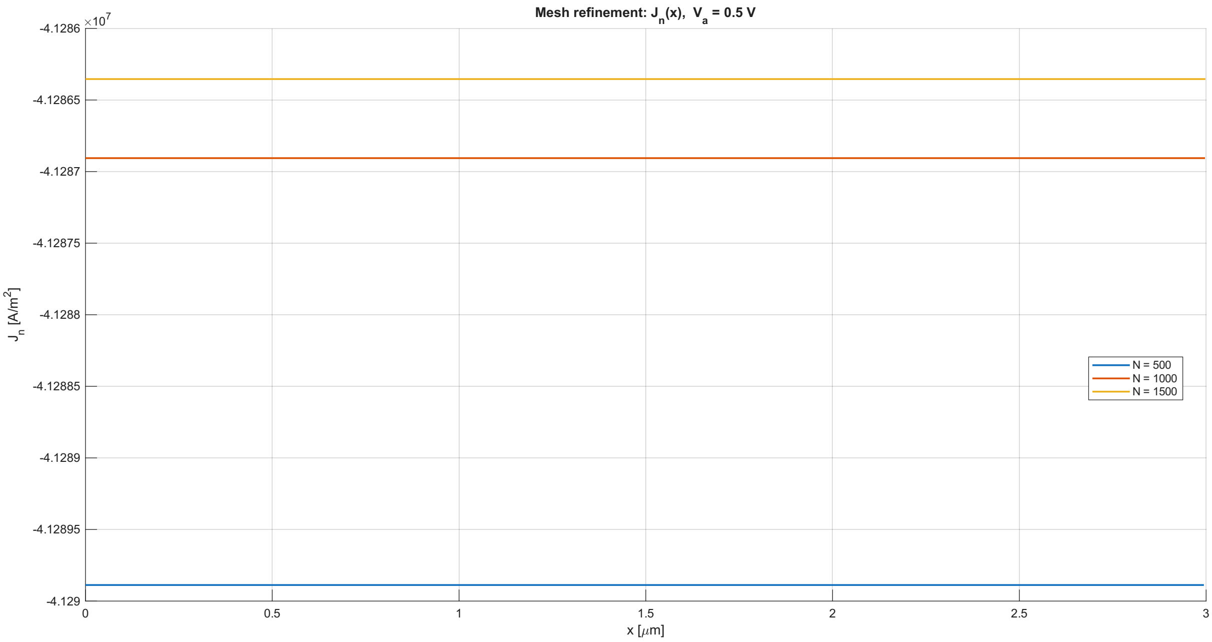

Figure 9: Mesh refinement study at $V_a = 0.5\,\mathrm{V}$: electron current density $J_n(x)$ for $N = 500$, 1000, and 1500 mesh points.

The results show that both the current density $J_n(x)$ and the maximum electron temperature $\max_x T_n$ are nearly identical across all three mesh sizes.
At $V_a = 0.5$ V, the variation in $\max T_n$ between $N = 500$ and $N = 1500$ is less than 0.01 K, and the current profiles are visually indistinguishable.

The residuals confirm that all three solutions are well converged: the Poisson residual remains at the level of 10^{-11}–10^{-12} and the global energy imbalance decreases monotonically with N. This indicates that even the coarsest mesh $N = 500$ provides sufficient spatial resolution to accurately describe the solution, including in the neighborhood of the n^+–n^- junctions where the doping profile varies most rapidly.

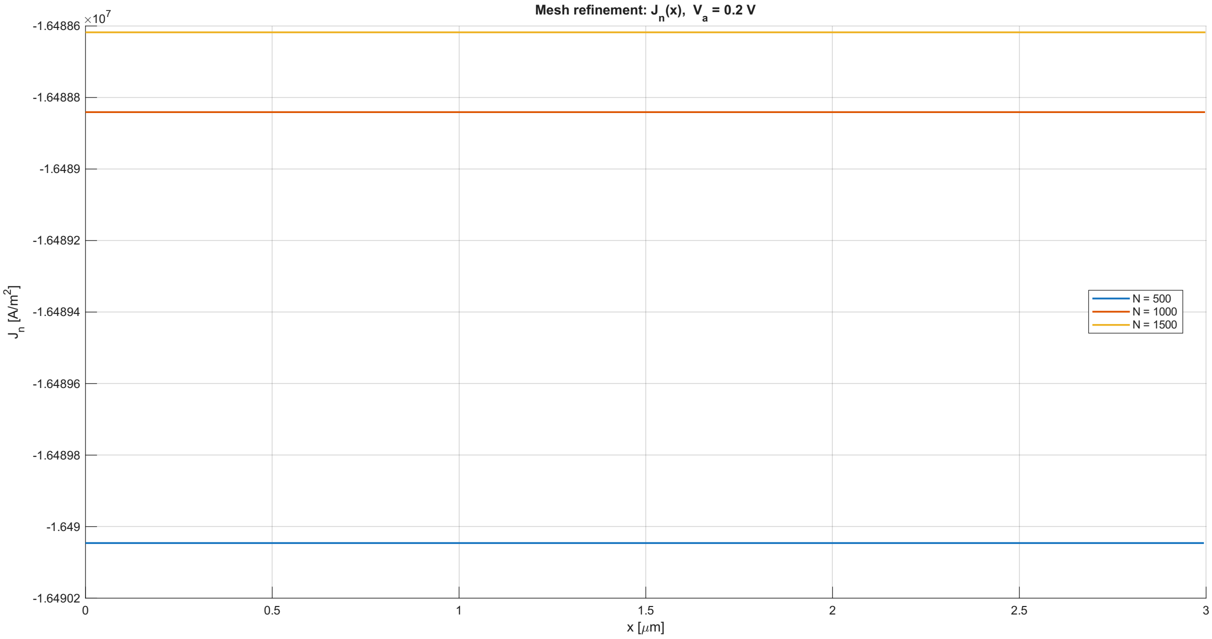

Figure 10: Mesh refinement study at $V_a = 0.2\,\mathrm{V}$: electron current density $J_n(x)$ for $N = 500$, 1000, and 1500 mesh points.

9 Point (10): Model validity and further discussion

9.1 Assumptions of the present model

The non-isothermal model adopted is an improvement with respect to the baseline isothermal drift-diffusion model. It relaxes the assumption of local thermal equilibrium between carriers and the lattice by introducing the electron temperature $T_n(x)$ as an additional unknown. However, it still relies on the following assumptions:

- **Local closure:** the current and energy flux are expressed through local quantities $(n,\ \varphi,\ \vartheta)$, implying that momentum and energy relaxation lengths are small compared to the device length $L = 3\,\mu\mathrm{m}$.
- **Constant mobility:** μ_n is independent of the electric field and of the electron temperature. This implies a linear drift-velocity law $u_n = \mu_n E$, which becomes inaccurate at high fields.
- **Continuum approximation:** carrier densities are smooth macroscopic fields; quantum and ballistic effects are neglected.

9.2 Validity indicators

The velocity ratio $\max_x |u_n|/v_{\mathrm{sat}}$ and the electron temperature rise $\max_x(T_n - T_L)/T_L$, reported in Table 4, provide quantitative indicators of how close the simulation is to the limits of the model. As discussed in Section 5, both indicators remain moderate across the bias range considered: at $V_a = 0.5$ V the velocity ratio reaches 0.26 and the temperature rise is approximately 3.7%. The model can therefore be considered physically credible for the device and bias range studied.

9.3 Improvements and new models

Here, some situations and improvements of the model are listed and shortly introduced:

- **Field-dependent mobility:** if $|u_n|/v_{\mathrm{sat}}$ approaches or exceeds 0.3, a velocity-saturation correction such as $\mu_n(E) = \frac{\mu_0}{(1+\mu_0|E|/v_{\mathrm{sat}})}$ should be introduced. This is the minimal extension and preserves the drift-diffusion structure.
- **Full hydrodynamic model:** the present energy-transport model neglects explicit momentum dynamics, assuming that momentum relaxation is fast. If inertial effects or velocity overshoot become

relevant — typically in sub-micron devices or at very high fields — a hydrodynamic formulation adding a momentum balance equation, thus computing the velocity of the carriers explicitly, may be required.

- **Boltzmann / Monte Carlo:** if the device length becomes comparable to the mean free path ($\mathrm{Kn} = \lambda/L \sim 1$), transport becomes quasi-ballistic and non-local. In this regime, continuum closures break down and a kinetic description based on the Boltzmann transport equation — solved stochastically via Monte Carlo — would be required. This regime is not reached for $L = 3\,\mu$m at the voltages considered.

The choice of which model to adopt is ultimately dictated by the physical regime of the device under study. In this work, for the device and bias range considered ($L = 3\,\mu$m, $V_a \leq 0.5$ V), the non-isothermal energy-balance model does not produce significant deviations from the isothermal drift-diffusion baseline: the relative difference in current density remains below 0.47% across the full bias sweep, and the maximum electron temperature rise is modest ($\approx 3.7\%$ above T_L at $V_a = 0.5$ V).

Nevertheless, the non-isothermal model provides a valuable diagnostic tool: by computing the electron temperature and the velocity ratio $\max_x |u_n|/v_{\mathrm{sat}}$, it offers a quantitative check on whether the isothermal drift-diffusion approximation is appropriate for the device under analysis. In this sense, the energy-balance model is not only a more accurate description of carrier transport, but also a principled way to assess the limits of the simpler baseline.

10 Appendix: codes

10.1 unipolar.m

```
clear; clc;% close all;

% Physical parameters and mesh
data;

%% Poisson matrix for -eps * psi''
%Nint is used because boundary have already been set
e = ones(Nint,1);
Apoisson = epsSi*spdiags([-e,2*e,-e],-1:1,Nint,Nint)/h^2;

%% Solver parameters
tol       = 1e-10;
maxit     = 500;

%relaxation parameters
omega_n    = 1;
omega_psi  = 1;
omega_theta = 1;

%% Storage

solutions = struct();
solutions_iso_Assignment = struct();
%% Step 1: Initial guess at first bias
Va = Va_list(1);

% Load of the isotermal baseline which are going to be used as initial
    guess
if isfile('isothermal_baseline.mat')
    load('isothermal_baseline.mat', 'solutions_iso');
    disp('Initial guess from isothermal solution correcly loaded!');
else
    error('isothermal_baseline.mat not found. Run isothermal case first!
        ');
```

```
end

if isfile('isothermal_baseline_Assignment.mat')
    load('isothermal_baseline_Assignment.mat', 'solutions_iso_Assignment
       ');
    disp('Baseline from isothermal solution correcly loaded!');
else
    error('isothermal_baseline.mat not found. Run isothermal case first!
       ');
end



% Boundary Conditions for the potential
psi_left  = 0;
psi_right = Va;

% Initial guess for the electron thermal voltage
theta = VT * ones(size(x));

% Boundary condition for the electron carrier density
n(1)   = n_left;
n(end) = n_right;

%% Bias continuation loop:
% Here the initialization can be choosen:
% Set init_strategy = 'I' (isothermal) or 'C' (continuation)
init_strategy = 'C';

for ibias = 1:length(Va_list)
    Va = Va_list(ibias);
    psi_left  = 0;
    psi_right = Va;
    fprintf('\n========================================\n');
    fprintf('Solving for Va = %.6g V\n', Va);
    fprintf('========================================\n');
    if init_strategy == 'I'
        %% Case I: Use of isothemal solutions for psi and n
        % Then bias continuation for theta
        % Initialization with the isotermal solution
        psi = solutions_iso(ibias).psi(:);
        psi(1)   = psi_left;
        psi(end) = psi_right;

        n = solutions_iso(ibias).n(:);

        Jn = solutions_iso(ibias).Jn(:); %this step is useless: just to
           be safe...

        if ibias == 1
            %initialized before
        else
            % First guess updated with the previous applied bias
            theta = solutions(ibias-1).theta;
        end

    elseif init_strategy == 'C'
        %% Case C: use of the previous bias loop as initial guess of the
```

```
        following one
    if ibias == 1
        psi = VT*log(ND/ND_high) + Va*x/L;

        psi(1)   = psi_left;
        psi(end) = psi_right;

        n = ND;
        n(1)   = n_left;
        n(end) = n_right;
        theta = VT * ones(size(x));
    else
        % Use previous solution as initial guess.
        n = solutions(ibias-1).n;
        n(1)   = n_left;
        n(end) = n_right;

        % Affine correction of potential to match the new voltage.
        Va_old = Va_list(ibias-1);
        dVa = Va - Va_old;

        psi = solutions(ibias-1).psi + dVa*x/L;
        psi(1)   = psi_left;
        psi(end) = psi_right;

        theta = solutions(ibias-1).theta;
        theta(1)   = theta_left;
        theta(end) = theta_right;
    end
end
err = 1 + tol;
it  = 0;

while err > tol && it < maxit

    it = it + 1;

    psi_old = psi;
    n_old   = n;

    %% Step 2: Continuity equation with Scharfetter-Gummel
    % find n
    n_new = continuity_n(psi, theta, mun, n_left, n_right, Nint,h);

    % Damping on density
    n = (1-omega_n)*n_old + omega_n*n_new;

    % Enforce density boundary values
    n(1)   = n_left;
    n(end) = n_right;

    %% Step 3: Jn calculation
    Jn = electron_current_sg(psi, n, h, theta, mun, q);

    %% Step 4: Solving of energy equation
    % Averages
    n_avg = (n(1:end-1)+n(2:end))/2;
    theta_avg = (theta(1:end-1) + theta(2:end))/2;
```

```
% lambda vector calculation
lambda = c_lambda*q*mun.*n_avg.*theta_avg;

theta_old = theta;

[theta_new, Sn, H] = energy_theta(psi, n, Jn, lambda, theta_left
    , theta_right, Nint, h, tau_E, VT, q);

theta = (1-omega_theta)*theta_old + omega_theta*theta_new;
%enforcing BC
theta(1) = theta_left;
theta(end) = theta_right;

%If the following line gets uncommentend, isothermal solutions
    are
%computed:
%theta = VT*ones(size(x));

%% 5. Gummel-linearized Poisson equation
% Applied linearization:
%
% n(psi^{k+1}) approx n^k + (n^k/VT)(psi^{k+1}-psi^k)
%
% Therefore:
%
% -eps psi'' + q n^k/VT psi
% =
% q(ND - n^k) + q n^k/VT psi^k
%From which the matrix have been extracted

n_int      = n(2:end-1);
ND_int     = ND(2:end-1);

psi_old_in = psi_old(2:end-1);

theta_int = theta(2:end-1);

Mreact = spdiags(q*n_int./theta_int,0,Nint,Nint);

%RHS
b = q*(ND_int - n_int) + q*(n_int./theta_int).*psi_old_in;

% Dirichlet boundary contributions
b(1)   = b(1)   + epsSi*psi_left/h^2;
b(end) = b(end) + epsSi*psi_right/h^2;

%computation on the new psi
psi_new = [psi_left; (Apoisson + Mreact)\b; psi_right];

% Damping on potential
psi = (1-omega_psi)*psi_old + omega_psi*psi_new;

% Enforce potential boundary values
psi(1)   = psi_left;
psi(end) = psi_right;

%% Step 7. Error indicators
```

```
    err_psi = norm(psi - psi_old,inf)/max([VT,norm(psi,inf)]);
    err_n   = norm(n - n_old,inf)/max(norm(n,inf),1);
    err_theta = norm(theta - theta_old, inf)/ norm(theta, inf);

    err = max([err_psi, err_n, err_theta]);

    fprintf('it = %4d | err = %.3e | err_psi = %.3e | err_n = %.3e |
        err_th = %.3e | min psi = %.4e | max psi = %.4e\n', ...
        it, err, err_psi, err_n, err_theta, min(psi), max(psi));

end

if it == maxit
    warning('Maximum number of iterations reached at Va = %.4g V',Va
       );
end

%% Save solution
solutions(ibias).Va    = Va;
solutions(ibias).psi   = psi;
solutions(ibias).n     = n;
solutions(ibias).Jn    = Jn;
solutions(ibias).theta = theta;



%% Plot current solution
figure(1); clf;

subplot(6,1,1);
semilogy(x*1e6,ND,'k--','LineWidth',1.2); hold on;
semilogy(x*1e6,n,'b','LineWidth',1.4);
semilogy(x*1e6, solutions_iso_Assignment(ibias).n, 'r','LineWidth'
   ,1.4);
grid on;
xlabel('x [\mum]');
ylabel('density [m^{-3}]');
legend('N_D', 'n (Non-Iso)', 'n (Iso)', 'Location', 'best');
title(sprintf('Electron density, Va = %.4g V',Va));

subplot(6,1,2);
plot(x*1e6,psi,'LineWidth',1.4); hold on;
plot(x*1e6, solutions_iso_Assignment(ibias).psi, 'r','LineWidth'
   ,1.4);
grid on;
xlabel('x [\mum]');
ylabel('\psi [V]');
legend('\psi (Non-Iso)', '\psi (Iso)', 'Location', 'best');
title('Electrostatic potential');

subplot(6,1,3);
Jn = solutions(ibias).Jn; hold on;
plot(x(1:end-1)*1e6,Jn,'LineWidth',1.4);
plot(x(1:end-1)*1e6, solutions_iso_Assignment(ibias).Jn, 'r','
   LineWidth',1.4);
grid on;
xlabel('x [\mum]');
```

```
ylabel('J_n [A/m^2]');
legend('J_n (Non-Iso)', 'J_n (Iso)', 'Location', 'best');
title('SG electron current density');

%Plot for electron temperature
Tn = theta * q / kB;

subplot(6,1,4)
plot(x*1e6, Tn, 'b', 'LineWidth', 1.4);
hold on;
yline(T, 'k--', 'T_L (Ambient)', 'LineWidth', 1.2, '
    LabelHorizontalAlignment', 'left');
grid on;
xlabel('x [\mum]');
ylabel('T_n [K]');
title(sprintf('Electron Temperature, Va = %.3g V', Va));

% % plot for theta
% subplot(6,1,4)
% plot(x*1e6,theta,'r','LineWidth',1.4);
% yline(VT, 'k--', 'V_T (Ambiente)', 'LineWidth', 1.2, '
    LabelHorizontalAlignment', 'left');
% grid on;
% hold on;
% xlabel('x [\mum]');
% ylabel('\vartheta [V]');
% title(sprintf('Electron Thermal Voltage, Va = %.3g V',Va));

subplot(6,1,5)
plot(x*1e6,n./ND,'LineWidth',1.4); hold on;
plot(x*1e6,(solutions_iso_Assignment(ibias).n)./ND,'LineWidth',1.4);
grid on;
xlabel('x [\mum]');
ylabel('n/N_D');
legend('n/N_D (ET)', 'n/N_D (Iso)', 'Location', 'best');
title(sprintf('Quasi-neutrality ratio, Va = %.3g V',Va));

subplot(6,1,6)
plot(x*1e6,q*(ND - n),'LineWidth',1.4); hold on;
plot(x*1e6,q*(ND - (solutions_iso_Assignment(ibias).n)),'LineWidth'
    ,1.4);
grid on;
xlabel('x [\mum]');
ylabel('\rho = q(N_D-n) [C/m^3]');
legend('\rho (Non-Iso)', '\rho (Iso)', 'Location', 'best');
title(sprintf('Space charge density, Va = %.3g V',Va));

drawnow;
%% Final residual diagnostics
fprintf('\nFinal diagnostics for Va = %.6e V\n', Va);

fprintf('min(psi)                     = %.6e V\n', min(psi));
fprintf('max(psi)                     = %.6e V\n', max(psi));

fprintf('min(n)                       = %.6e m^-3\n', min(n));
fprintf('max(n)                       = %.6e m^-3\n', max(n));
fprintf('min(n/ND)                    = %.6e\n', min(n./ND));
fprintf('max(n/ND)                    = %.6e\n', max(n./ND));
```

```
fprintf('mean(Jn)                    = %.6e A/m^2\n', mean(Jn));
fprintf('min(Jn)                     = %.6e A/m^2\n', min(Jn));
fprintf('max(Jn)                     = %.6e A/m^2\n', max(Jn));

% --------- Current conservation diagnostic -----------%
%Jn is computed again because at the end of the Gummel iterations
   the
%last theta have to be taken into account
Jn = electron_current_sg(psi, n, h, theta, mun, q);
Jspread = max(Jn) - min(Jn);

Jscale = q*mun*VT*ND_high/L;

rel_scale   = Jspread / Jscale;

fprintf('\n--- Diagnostic 11.1: Current conservation ---\n');
fprintf('Jscale                          = %.6e A/m^2\n', Jscale);
fprintf('spread(Jn)                      = %.6e A/m^2\n', Jspread);
fprintf('spread(Jn) / Jscale             = %.6e\n', rel_scale);

% This ratio is meaning full only far form the equilibrium
if abs(mean(Jn)) > 1e-12 * Jscale
    rel_mean = Jspread / abs(mean(Jn));
    fprintf('spread(Jn) / |mean(Jn)|         = %.6e\n', rel_mean);
else
    fprintf('spread(Jn) / |mean(Jn)|         = not significative,
       current is almost nil\n');
end

%-------------- Poisson residual Diagnostic ---------------%

rho_int = q*(ND(2:end-1) - n(2:end-1));

lhsP = -epsSi*(psi(3:end)-2*psi(2:end-1)+psi(1:end-2))/h^2;
rhsP = rho_int;

resP = lhsP - rhsP;

Pscale = q*ND_high;
fprintf('\n--- Diagnostics 11.2: Poisson Residual ---\n');
fprintf('Poisson residual inf       = %.6e C/m^3\n', norm(resP,inf));
fprintf('Poisson residual / Pscale = %.6e\n', norm(resP,inf)/Pscale)
   ;
fprintf('Poisson residual / max(abs(rho_int)) = %.6e\n', ...
    norm(resP,inf)/max(norm(rho_int,inf),1));

%----------------- Energy residual Diagnostic -------------%

alpha = 1.5*q*n(2:end-1)/tau_E;

dSn = (Sn(2:end)-Sn(1:end-1))/h;

resEN = dSn - (H - alpha.*(theta_int - VT));

%denom_EN_RES = max(abs(H) + abs(alpha.*(theta_int - VT))) + eps;
denom_EN_RES = max(abs(H + alpha.*(theta_int - VT))) + eps;
```

```
fprintf('\n--- Diagnostics 11.3: Energy residual ---\n')
fprintf('||R^E||_inf / denom            = %.6e\n', norm(resEN, inf) /
    denom_EN_RES);

% ---------------- Global Energy Balance Diagnostic
   ----------------%

[~, ~, imbalance] = energy_balance_diagnostics(psi, n, theta, Jn, Sn
   , q, VT, tau_E, h);

fprintf('\n--- Diagnostics 11.4: Global energy balance ---\n');
fprintf('Relative imbalance             = %.6e\n', imbalance);

%----------------- Drift Velocity Diagnostics --------------%

vsat = 1e5;   % m/s, typical electron saturation velocity in Si
n_edge = 0.5*(n(1:end-1) + n(2:end));

% Actual carrier mean velocity
u_edge = -Jn./(q*n_edge);
fprintf('\n--- Diagnostics 11.5: Velocity and Validity indicators
   ---\n');

fprintf('max(|u_n|)              = %.6e m/s\n', max(abs(u_edge)));
fprintf('max(|u_n|)/vsat         = %.6e\n', max(abs(u_edge))/vsat);

ratio_vsat = max(abs(u_edge))/vsat;
if ratio_vsat < 0.2
    fprintf('Regime: low-field, mobility model valid\n');
elseif ratio_vsat < 0.3
    fprintf('Regime: acceptable \n');
else
    fprintf('Regime: WARNING - constant mobility questionable\n');
end

%----------------- Quasi-Fermi diagnostics --------------%
%it is applied for the isothermal case: may be useful to compare
   with
% velocity drift checks
psi_iso = solutions_iso(ibias).psi;
n_iso   = solutions_iso(ibias).n;

psiF  = psi_iso - VT * log(n_iso / ND_high);
Fqf   = -diff(psiF) / h;
u_qf  = mun * abs(Fqf);

fprintf('max(mu*|grad psiF|)/vsat = %.6e  [isothermic]\n', max(u_qf)
   /vsat);

%---------------Temperature Checks -------------------------%

% Electron temperature
Tn = theta * q / kB;

Tn_max = max(Tn);
Tn_min = min(Tn);
dT_rel_max = max((Tn-T)/T);
```

```
    fprintf('\n--- Diagnostics 11.6: Temperature checks ---\n');
    fprintf('max(Tn)                              = %.4f K\n', Tn_max);
    fprintf('min(Tn)                              = %.4f K\n', Tn_min);
    fprintf('max((Tn-TL)/TL)                      = %.6e\n',  dT_rel_max);

    if Tn_min <= 0
        error('The temperature is negative!');
    end
    %keyboard;
end
%------------- Report on carrier velocity for every bias
   ----------------%
fprintf('\n--- Report on carrier velocity for every bias ---\n');
fprintf('%-10s %-15s %-15s\n', 'Va [V]', 'max|u|/vsat', 'regime');
for ibias = 1:length(solutions)
    Va    = solutions(ibias).Va;
    Jn    = solutions(ibias).Jn;
    n     = solutions(ibias).n;

    n_edge = 0.5*(n(1:end-1)+n(2:end));
    u_edge = -Jn./(q*n_edge);

    ratio  = max(abs(u_edge))/vsat;
    if ratio < 0.2,       regime = 'low-field';
    elseif ratio < 0.3,   regime = 'acceptable';
    else,                 regime = 'WARNING';
    end
    fprintf('%-10.3f %-15.3e %-15s\n', Va, ratio, regime);
end

%% I-V characteristics
Va_values = arrayfun(@(s) s.Va, solutions);
J_values  = arrayfun(@(s) mean(s.Jn), solutions);

J_iso_values = arrayfun(@(s) mean(s.Jn), solutions_iso_Assignment);

figure(2); clf;
plot(Va_values,J_values, 'o-r','LineWidth',1.4);
hold on;
plot(Va_values,J_iso_values,'o-b','LineWidth',1.4);
grid on;
xlabel('V_a [V]');
ylabel('mean J_n [A/m^2]');
legend('non_isothermal', 'Isothermal', 'Location', 'best');
title('Current-voltage characteristic');

%% Current comparison: Non-Isothermal vs Isotermal (Va = 0 excluded)
% Length extraction
N_sim = length(solutions);


if N_sim > 1
    %I ignore Va=0
    Va_values     = arrayfun(@(s) s.Va, solutions(2:N_sim));
    J_values      = arrayfun(@(s) mean(s.Jn), solutions(2:N_sim));
    J_iso_values = arrayfun(@(s) mean(s.Jn), solutions_iso_Assignment(2:
       N_sim));
```

```
    % absolute difference computation
    Delta_J = J_values - J_iso_values;

    % relative difference computation
    Delta_J_perc = zeros(1, length(Va_values));
    idx_nz = abs(J_iso_values) > 1e-12;
    Delta_J_perc(idx_nz) = (Delta_J(idx_nz) ./ abs(J_iso_values(idx_nz))
        ) * 100;

    %Plot
    figure(3); clf;
    set(gcf, 'Name', 'Comparison ET vs DD');

    % Subplot 1: absolute difference
    subplot(2,1,1);
    plot(Va_values, Delta_J, 'o-', 'LineWidth', 1.5, 'Color', '#D95319',
         'MarkerFaceColor', '#D95319');
    grid on;
    xlabel('V_a [V]');
    ylabel('\Delta J_n [A/m^2]');
    title('Absolute Difference: J_{n, ET} - J_{n, DD} (V_a > 0)');

    % Subplot 2: relative difference
    subplot(2,1,2);
    plot(Va_values, Delta_J_perc, 's-', 'LineWidth', 1.5, 'Color', '
        #0072BD', 'MarkerFaceColor', '#0072BD');
    grid on;
    xlabel('V_a [V]');
    ylabel('\Delta J_n / J_{n, DD} [%]');
    title('Relative variation of current (V_a > 0)');
 else
     disp('Warning: only Va = 0 have been computed. No comparison
         available');
 end

%% Figure 4: Mesh refinement (persistent across runs, single bias)
ibias = 7;
Va_target = Va_list(ibias);
Jn = solutions(ibias).Jn;

figure(4);
hold on;
plot(x(1:end-1)*1e6, Jn, 'LineWidth', 1.4, ...
     'DisplayName', sprintf('N = %d', N));
grid on;
xlabel('x [\mum]');
ylabel('J_n [A/m^2]');
title(sprintf('Mesh refinement: J_n(x),  V_a = %.2g V', Va_target));
legend('Location', 'best');
drawnow;

ibias = 4;
Va_target = Va_list(ibias);
Jn = solutions(ibias).Jn;

figure(5);
hold on;
plot(x(1:end-1)*1e6, Jn, 'LineWidth', 1.4, ...
```

```
    'DisplayName', sprintf('N = %d', N));
grid on;
xlabel('x [\mum]');
ylabel('J_n [A/m^2]');
title(sprintf('Mesh refinement: J_n(x),  V_a = %.2g V', Va_target));
legend('Location', 'best');
drawnow;
```

10.2 log_mean.m

```
    function lm = log_mean(a,b)
%log_mean computed the logarithmic mean between two vectors a & b
a=a(:);
b=b(:);

% lm gets initialized at the limit case in which a=b

lm = a;

tol = 1e-8;

%The cases in which b != a get distinguished and the updated according
    to
%the formula
idx = abs(b-a) > tol;

lm(idx) = (b(idx) - a(idx)) ./ log(b(idx)./a(idx));

end
```

10.3 energy_theta.m

```
function [theta, Sn, H] = energy_theta(psi, n, Jn, lambda, theta_a,
    theta_b, Nint, h, tau_E, VT, q)

% Peclet number

PE = (5*Jn*h)./(2*lambda);
[bp,bn] = bern(PE);

% %electric field computation
 E = -(psi(2:end)-psi(1:end-1))/h;
% E1 = -diff(psi)/h;


% H coefficient of the equation
H = 0.5*(Jn(2:end).*E(2:end) + Jn(1:end-1).*E(1:end-1));

% alpha coefficient: only internal node of n are used for alpha vector
alpha = 1.5*(q*n(2:end-1))/tau_E;

diag_main_en = (lambda(2:end).*bn(2:end)+lambda(1:end-1).*bp(1:end-1))/h
    ;
diag_main = diag_main_en + h*alpha;

% Subdiagonal: row r couples to node r-1
% Valid for r = 2,...,Nint
```

```
rows_low = (2:Nint).';
cols_low = (1:Nint-1).';
vals_low = -(lambda(2:Nint).*bn(2:Nint))/h ;

% Superdiagonal: row r couples to node r+1
% Valid for r = 1,...,Nint-1
rows_up = (1:Nint-1).';
cols_up = (2:Nint).';
vals_up = -(lambda(2:Nint).*bp(2:Nint))/h;

% Main diagonal
rows_diag = (1:Nint).';
cols_diag = (1:Nint).';
vals_diag = diag_main(:);

% Assemble matrix
rows = [rows_diag; rows_low; rows_up];
cols = [cols_diag; cols_low; cols_up];
vals = [vals_diag; vals_low; vals_up];

A = sparse(rows,cols,vals,Nint,Nint);

%RHS of the equation
rhs = h*(H+alpha*VT);


A_low_boundary = -(lambda(1)*bn(1))/h;
A_up_boundary  = -(lambda(end)*bp(end))/h;

% I must include external nodes (Boundary conditions)
rhs(1)   = rhs(1)   - A_low_boundary * theta_a;
rhs(end) = rhs(end) - A_up_boundary  * theta_b;

% Solve the linear system for theta
theta_int = A \ rhs;

theta = [theta_a; theta_int; theta_b];

%%For diagnostic 11.3
Sn = (lambda)/h .* (theta(1:end-1).*bn-theta(2:end).*bp);

if any(theta < 0)
        error('!!! Negative thetha !!!');
end
```

10.4 energy_balance_diagnostics.m

```
function [lhs, relax_term, imbalance] = energy_balance_diagnostics(psi,
   n, theta, Jn, Sn, q, VT, tau_E, h)
    % LHS integral
    E = -(psi(2:end) - psi(1:end-1)) / h;
    JnE = Jn .* E;

    lhs = (JnE(1)/2+ sum(JnE(2:end-1)) + JnE(end)/2)*h;

    %RHS
    relax_term = 1.5*q*n(2:end-1).*(theta(2:end-1) - VT)/tau_E;
```

```
    int_relax_term = h*(relax_term(1)/2 + ...
        sum(relax_term(2:end-1)) + relax_term(end)/2);

    rhs = Sn(end) - Sn(1) + int_relax_term;

    %Identity check: eps is added in case LHS = 0 as for Va = 0
    imbalance = abs(lhs - rhs) / (abs(lhs) + eps);
```

10.5 electron_current_sg.m

```
function Jn = electron_current_sg(psi, n, h, theta, mun, q)

%Computation of the logarithmic mean of theta:

theta_log_mean = log_mean(theta(2:end), theta(1:end-1));

eta = (diff(psi) - diff(theta)) ./ theta_log_mean;

Dn = mun*theta_log_mean;

[bp,bn] = bern(eta);

Jn = q*Dn/h.*(n(2:end).*bp - n(1:end-1).*bn );

end
```

10.6 data.m

```
%%This script works as a pre-processing to the method application

%% Physical constants
q     = 1.602176634e-19;       % C
kB    = 1.380649e-23;          % J/K
T     = 300;                   % K
%In the project I will include the impact of hot electrons
VT    = kB*T/q;                % thermal voltage, V

c_lambda = 5/2;

%tau_E = [0.05, 0.1, 0.2, 0.5, 1.0]*1e-12
tau_E = 1e-12;

eps0   = 8.8541878128e-12;     % F/m
epsSi  = 11.7*eps0;            % F/m

%% Device geometry
% L      = 1e-6;                 % m
% Lc     = 0.15e-6;              % m
% sigma = 10e-9;                 % m

L      = 3e-6;
Lc     = 0.45e-6; %thickness of high doped region
sigma = 30e-9; %this modulates the htan of the doping

%% Doping: 100 difference
ND_low   = 1e22;               % m^-3 = 1e16 cm^-3
ND_high = 1e24;                % m^-3 = 1e18 cm^-3
```

```
%% Electron mobility and diffusion coefficient
%also in the project mobility is constant
mun = 0.10;                     % m^2/(V s)
%NB: Einstein does not hold for differente temperatures
Dn  = mun*VT;                   % m^2/s

%% Mesh-> uniform
N = 1000;                        % total number of nodes
x = linspace(0,L,N).';
h = x(2)-x(1);

Nint = N-2;                     % number of interior nodes

%% Smooth n+-n-n+ doping profile
% As given by the exercise
%First the slope: small sigma means more "step-like" doping, they are
%already shifted on the x axis
SL = 0.5*(1 - tanh((x - Lc)/sigma));
SR = 0.5*(1 + tanh((x - (L-Lc))/sigma));

ND = ND_low + (ND_high - ND_low).*(SL + SR);

%% Boundary conditions for density
n_left  = ND_high;
n_right = ND_high;

%% Boundary condition for electron thermal voltage theta
theta_left = VT;
theta_right = VT;

%% Bias continuation values
Va_list = [0, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5];  % V
%the device is symmetrical, thus i may apply negative voltages as well
%this set is used for the characteristic: the previous value is the
   initial
%guess of the next one

%Va_list = linspace(0,0.5,20);

% Va_list = 0;          % use this for equilibrium only

%% Useful scales
LD_low  = sqrt(epsSi*VT/(q*ND_low));
LD_high = sqrt(epsSi*VT/(q*ND_high));

%Voltage difference at equilibrium between high doped lateral zone and
   low
% doped central zone
Vbi     = VT*log(ND_high/ND_low);

Jscale = (q*mun*VT*ND_high)/L;
fprintf('VT      = %.4g V\n', VT);
fprintf('Dn      = %.4g m^2/s\n', Dn);
fprintf('LD_low  = %.4g nm\n', LD_low*1e9);
fprintf('LD_high = %.4g nm\n', LD_high*1e9);
fprintf('Vbi     = %.4g V\n', Vbi);
```

10.7 continuity_n.m

```
function n = continuity_n(psi, theta, mun, na, nb, Nint, h)
%% Used to solve current continuity equation

% Nint = number of interior nodes
% total number of nodes is Nint+2

psi   = psi(:);
theta = theta(:);
%Computation of the logarithmic mean of theta:
theta_log_mean = log_mean(theta(2:end), theta(1:end-1));

% theta = diff(psi)/Vth;

eta = (diff(psi) - diff(theta)) ./ theta_log_mean;

% Convention:
% bp(i) = B(eta_i)
% bn(i) = B(-eta_i)
[bp,bn] = bern(eta);

Dn = mun*theta_log_mean;
%c is a vector in the non-isothermal case
c = Dn/h;

% Diagonal entries for interior nodes
diag_main = c(1:end-1).*bp(1:end-1) + c(2:end).*bn(2:end);

% Subdiagonal: row r couples to node r-1
% Valid for r = 2,...,Nint
rows_low = (2:Nint).';
cols_low = (1:Nint-1).';
vals_low = -c(2:end-1).*bn(2:end-1);

% Superdiagonal: row r couples to node r+1
% Valid for r = 1,...,Nint-1
rows_up = (1:Nint-1).';
cols_up = (2:Nint).';
vals_up = -c(2:end-1).*bp(2:end-1);

% Main diagonal
rows_diag = (1:Nint).';
cols_diag = (1:Nint).';
vals_diag = diag_main(:);

% Assemble matrix
rows = [rows_diag; rows_low; rows_up];
cols = [cols_diag; cols_low; cols_up];
vals = [vals_diag; vals_low; vals_up];

An = sparse(rows,cols,vals,Nint,Nint);

% Boundary contributions
Fn = sparse(Nint,1);
Fn(1)   = c(1)*bn(1)*na;
Fn(end) = c(end)*bp(end)*nb;
```

```
n_int = An\Fn;

n = [na; n_int; nb];

end
```

10.8 Bernoulli functions

10.8.1 bernm̀

```
function [bp,bn] = bern(theta)

[bp,bn] = bern_mike(theta);
% bp = BernoulliF(theta);
% bn = BernoulliF(-theta);
end
```

10.8.2 BernoulliF.m

```
function B = BernoulliF(z)

B = zeros(size(z));

small = abs(z) < 1e-8;
large = ~small;

% Series expansion near zero:
% B(z) = z/(exp(z)-1)
%      = 1 - z/2 + z^2/12 - z^4/720 + ...
zs = z(small);
B(small) = 1 - zs/2 + zs.^2/12 - zs.^4/720;

zl = z(large);
B(large) = zl ./ expm1(zl);

end
```

10.8.3 bern_mike.m

```
    function [bp,bn] = bern_mike(x)

% Vector implementation of the Bernoulli function.
% Returns bp = B(x) and bn = B(-x) for an input vector x.
%
% Observations:
%   for x >  709, exp( x) = inf ==> b returns  0
%   for x < -745, exp(-x) = 0   ==> b returns -x
%
% Properties:
%   bn = exp(x)*bp;
%   bn = x + bp;
%
%   Points of strength:
%   - ensures bp(x) = bn(-x) for every x
%   - precision in on the order of epsM
%   - xlim = 2^-2 ensures maximum precision for every x
%     at least for Taylor to the 10   order (trial-and-error)
```

```
%

bp = zeros(size(x));
bn = bp;
xlim = 2^-2;

% |x| big & x positive
ibig         =  find(x > xlim);
if ~isempty(ibig)
  bp(ibig)    =  x(ibig)./(exp(x(ibig))-1);
  bn(ibig)    =  x(ibig) + bp(ibig);
end

% |x| big & x negative
ibig         =  find(x < xlim);
if ~isempty(ibig)
  bn(ibig)    = -x(ibig)./(exp(-x(ibig))-1);
  bp(ibig)    =  bn(ibig) - x(ibig);
end

% |x| small
ismall       = find(abs(x) <= xlim);
if ~isempty(ismall)
  bp(ismall) = x(ismall).^10/47900160 - x(ismall).^8/1209600 + ...
               x(ismall).^6 /30240     - x(ismall).^4/720     + ...
               x(ismall).^2 /12        - x(ismall)   /2       + 1;

  bn(ismall) = x(ismall).^10/47900160 - x(ismall).^8/1209600 + ...
               x(ismall).^6 /30240     - x(ismall).^4/720     + ...
               x(ismall).^2 /12        + x(ismall)   /2       + 1;
end
```